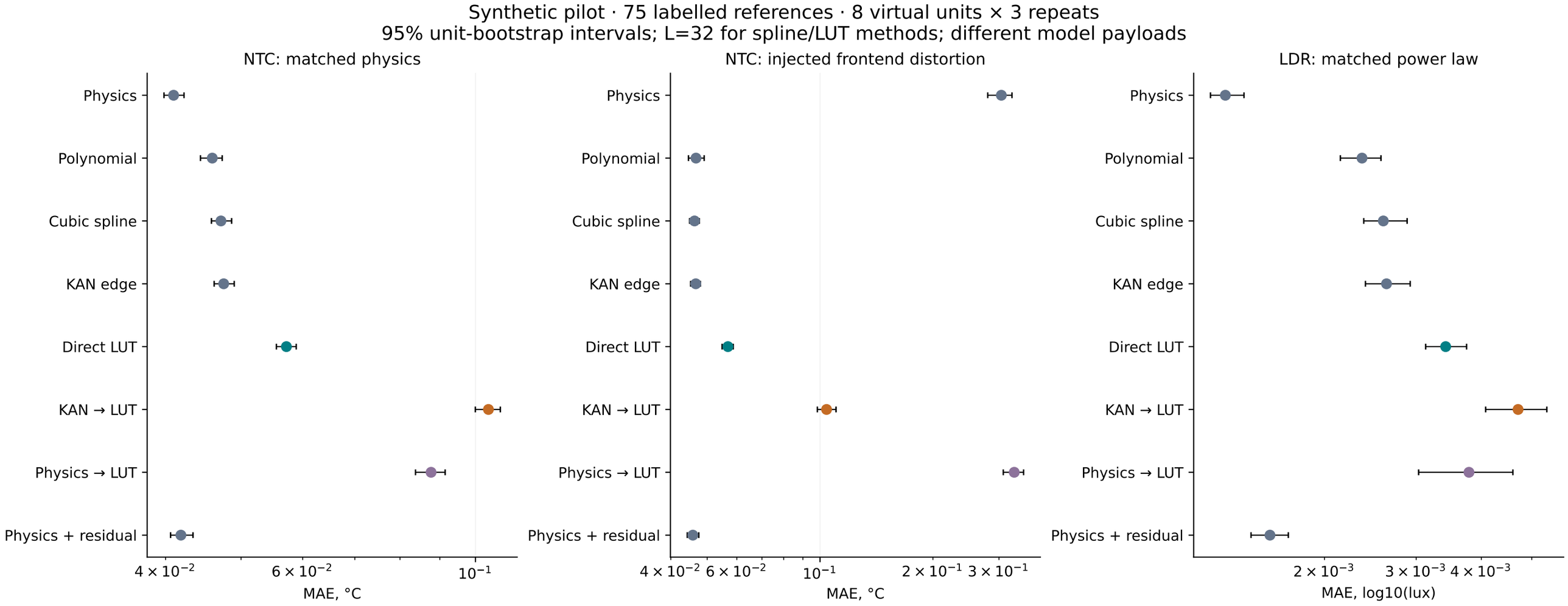
Synthetic pilot · 75 labelled references · 8 virtual units × 3 repeats
95% unit-bootstrap intervals; L=32 for spline/LUT methods; different model payloads
NTC: matched physics
NTC: injected frontend distortion
LDR: matched power law
Physics
Polynomial
Cubic spline
KAN edge
Direct LUT
KAN → LUT
Physics → LUT
Physics + residual
4 × 10⁻²
6 × 10⁻²
10⁻¹
2 × 10⁻¹
3 × 10⁻¹
2 × 10⁻³
3 × 10⁻³
4 × 10⁻³
MAE, °C
MAE, °C
MAE, log10(lux)